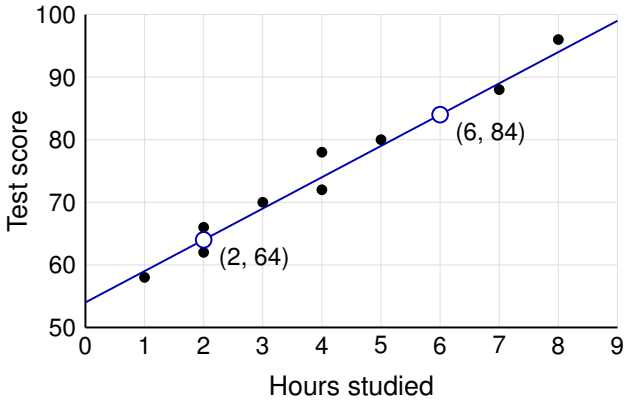
Scatterplots and Lines of Best Fit KEY

Name _____

Ten students recorded how many hours they studied and their test score. A **scatterplot** shows both numbers for each student. Each dot is one student.



Direction	**Positive**: as x goes up, y goes up. (Negative: y goes down. No pattern at all: no association.)
Form	**Linear**: the dots follow a line. (Or curved.)
Strength	**Strong**: the dots are close to the line. (Or moderate: a little spread out. Or weak: very spread out.)
Unusual	Outliers or clusters? None here.

Line of best fit. A line that follows the trend of the dots.

- It goes through the middle of the dots.
- About the same number of dots are above it as below it.

Finding its equation. Pick two points **on the line** (they do not have to be dots): $(2, 64)$ and $(6, 84)$.

Slope: $m = \dfrac{84 - 64}{6 - 2} = \dfrac{20}{4} = 5$

y-intercept: $64 = 5(2) + b$, so $b = 54$

Equation: $\hat{y} = 5x + 54$

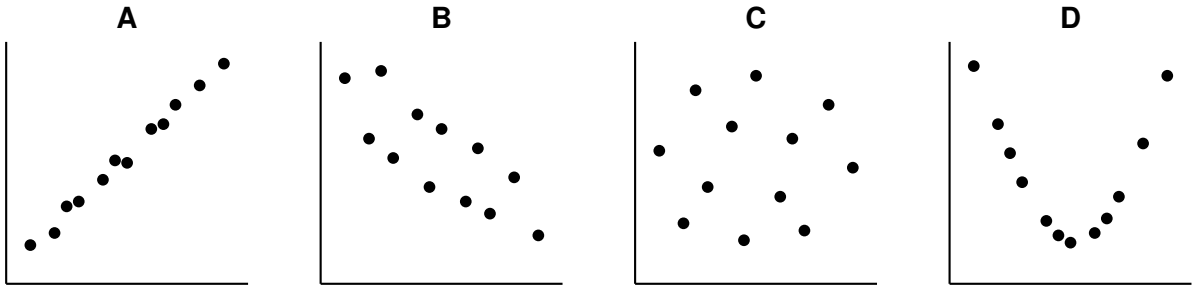
$\hat{y}$ is read "y-hat." The hat means **predicted**. The line predicts y; it does not give the exact y.

Predicting. A student studies 5 hours: $\hat{y} = 5(5) + 54 = 79$. The predicted score is 79.

Gotcha: extrapolation. Predicting far outside the data is risky. 12 hours gives $\hat{y} = 5(12) + 54 = 114$, but a test score cannot be more than 100. The data only goes from 1 to 8 hours.

Work with a partner. Compare answers when you finish each problem.

1. Describe the direction, form, and strength of each scatterplot.



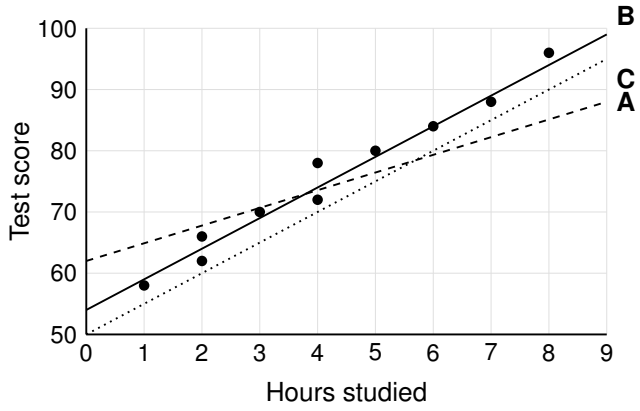
(a) A Positive, linear, strong

(b) B Negative, linear, moderate (the dots are spread out from a line)

(c) C No association: no direction or pattern

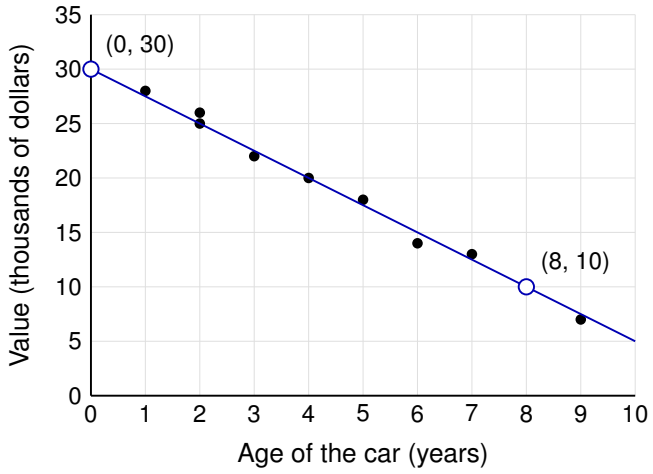
(d) D Curved (not linear), strong. It goes down and then up.

2. Which line fits the study data best? Explain why the other two do not.



Line B. It follows the trend, with about as many dots above as below. Line A is too flat: it is above the low dots and below the high dots. Line C is below all the dots.

3. This scatterplot shows the age and value of 10 used cars. A line of best fit is drawn.



(a) Describe the direction, form, and strength. Negative, linear, strong

(b) Use the two marked points to find the slope.
$m = \frac{10 - 30}{8 - 0} = \frac{-20}{8} = -2.5$

(c) Write the equation of the line. $\hat{y} = -2.5x + 30$

(d) Predict the value of a 6-year-old car.
$\hat{y} = -2.5(6) + 30 = 15$, about \$15,000

(e) Predict the value of a 15-year-old car. Does your answer make sense? Explain.
$\hat{y} = -2.5(15) + 30 = -7.5$. No. A car cannot be worth less than \$0. 15 years is far outside the data (1 to 9 years).

4. The line for the study data is $\hat{y} = 5x + 54$.

(a) Predict the score of a student who studies 3 hours. $5(3) + 54 = 69$

(b) A student plans to study 10 hours. Predict the score. Should you trust this prediction? Explain.
$5(10) + 54 = 104$. No. Scores cannot be more than 100, and 10 hours is outside the data.